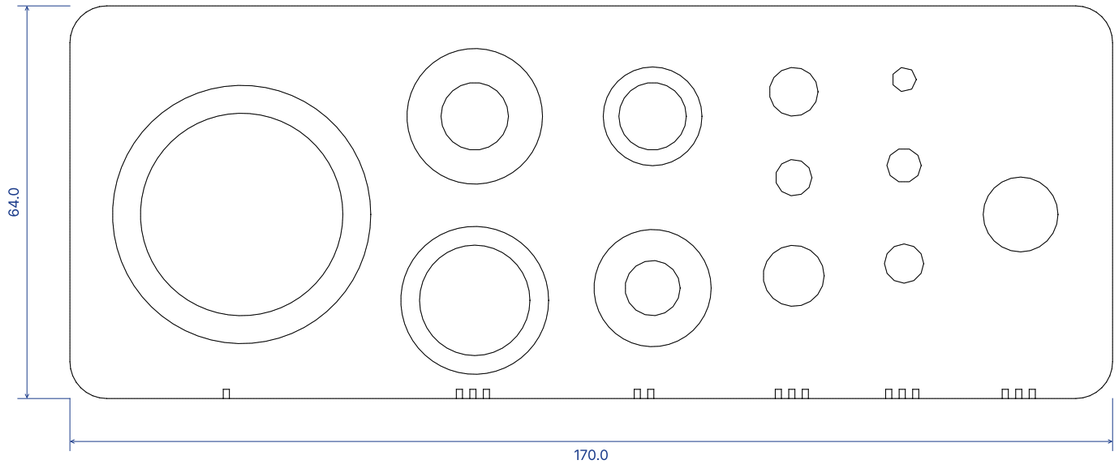
WRA-1 wheelchair robotic arm - part drawings

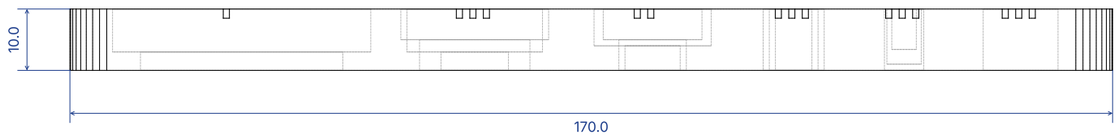
Generated from the parametric source (cad/wra). Units mm. Third-angle projection.
Dimensions shown are overall sizes and auto-extracted bore tables; the STEP files are the master geometry.

1	00_fit_test_coupon	x1	same as production	PRINT FIRST: fit-test coupon (bearings, pins, inserts)
2	01_receiver	x1	PETG (PA-CF optional)	Dock receiver (chair side, same part L/R)
3	02_tube_clamp_body	x5	PA-CF recommended (PETG ok)	Tube clamp body (parametric tube OD)
4	03_tube_clamp_cap	x5	PA-CF recommended (PETG ok)	Tube clamp cap
5	04_post_bracket	x1	PA-CF recommended	Post bracket (spine + shelf), same part L/R
6	05_stop_collar	x1	PETG	Post stop collar (secondary retention)
7	10_base_housing	x1	PA-CF (required: dock plate	Base housing + dock plate
8	11_j1_fixed_pulley	x1	PETG	J1 fixed pulley 54T (+ hard-stop arc)
9	12_dm_bracket	x1	PETG	Drive-module bracket
10	12_dm_cap	x3	PETG	Drive-module servo cap
11	12_dm_pulley36	x3	PETG	36T HTD-3M drive pulley
12	20_turntable	x1	PETG (PA-CF optional)	Turntable plate + J1 spindle
13	21_stub_cheek	x1	PETG (PA-CF recommended)	Stub cheek (clamps J2 stub axle)
14	22_ddm_bracket_j2	x1	PETG	Dual drive bracket (J2 half)
15	22_ddm_bracket_j3	x1	PETG	Dual drive bracket (J3 half)
16	30_ua_hub	x1	PETG (PA-CF recommended)	Upper-arm hub with integral 86T (J2) pulley
17	31_j3_hub	x1	PETG (PA-CF recommended)	J3 hub: 72T pulley + parallelogram crank + spring L-lever
18	32_elbow_block	x1	PETG (PA-CF recommended)	Elbow block (clamps elbow pin, J2 spring anchor, stop lugs)
19	33_rod_end	x2	PETG	Parallelogram rod end (626 bearing)
20	34_forearm_root	x1	PETG (PA-CF recommended)	Forearm root: elbow bearings + crank + stop tab
21	35_stub_spacer_14p46mm	x1	PETG	Stub inner-ring spacer 14.46 mm
22	35_stub_spacer_30p96mm	x1	PETG	Stub inner-ring spacer 30.96 mm
23	40_wrist_housing	x1	PETG (PA-CF recommended)	Wrist housing: forearm sockets + J4 drive cradle + pitch for
24	41_pitch_yoke	x1	PETG (PA-CF recommended)	Pitch yoke: J5 servo pocket + roll bearing housing
25	42_yoke_cap	x1	PETG	Pitch-yoke rear cap
26	43_j4_pulley54	x1	PETG	J4 54T pulley (bolts to yoke)
27	44_j5_horn_adapter	x1	PETG (PA-CF)	J5 horn adapter with drive dogs
28	45_roll_spindle	x1	PA-CF or PETG	Roll spindle
29	46_tool_flange	x1	PETG (PA-CF)	Tool flange (WRA-TF30 interface)
30	47_roll_spacer	x1	PETG	Roll-spindle inner-ring spacer
31	50_claw_palm	x1	PA-CF (PETG ok)	Claw palm: dovetail, servo cradle, rail beam
32	51_claw_cradle_cap	x1	PETG	Claw servo cradle cap
33	52_claw_pinion	x1	PA-CF / PETG-CF	Claw pinion 14T m1.5 (on servo horn)
34	53_claw_jaw_a	x1	PA-CF / PETG-CF	Claw jaw A (rack on top of pinion, upper knob lip)
35	53_claw_jaw_b	x1	PA-CF / PETG-CF	Claw jaw B (rack below pinion, lower knob lip)
36	54_claw_pad_a	x1	TPU 95A + silicone sheet	Claw pad A, 100 deg V (glue 1.5 mm silicone facing)
37	54_claw_pad_b	x1	TPU 95A + silicone sheet	Claw pad B (mirror of A)
38	56_tool_adapter	x1	PA-CF (PETG ok)	Quick-change tool adapter (dovetail, both wrists)
39	60_electronics_box	x1	PETG	Electronics enclosure (Jetson, ESP32, power, E-stop relays)
40	61_electronics_lid	x1	PETG	Electronics enclosure lid
41	62_estop_pendant	x1	PETG (yellow recommended)	E-stop pendant body
42	63_estop_pendant_lid	x1	PETG (yellow)	E-stop pendant lid (E-stop + guarded START)
43	70_bench_test_stand	x1	PETG	Bench test stand (holds a post tube like the chair)
44	80_cs_gear_front	x1	PA-CF (PETG-CF ok)	CS-gear front hemisphere + tool flange
45	81_cs_gear_back	x1	PA-CF (PETG-CF ok)	CS-gear back hemisphere
46	82_monopole	x2	PA-CF	Monopole S-gear 17T m1.5 + 15T m1.5 drive gear
47	83_mp_fork	x2	PA-CF (PETG-CF ok)	Monopole fork: pin arms + 6808 inner-race ring (carrier part
48	84_mp_cradle	x2	PA-CF (PETG ok)	Theta-servo cradle: sleeve, U-channel, 6805 hub, q dogs (car
49	85_theta_gear	x2	PA-CF / PETG-CF	Theta drive gear 20T m1.5 (on servo horn)
50	86_q_coupling	x2	PA-CF / PETG	q-servo horn coupling with drive dogs
51	87_module_frame	x2	PA-CF (PETG ok)	Module frame: 6808 holder, lug, strut, 6805 boss, q-servo cr
52	88_q_servo_cap	x2	PETG	q-servo cap
53	89_abenics_housing	x1	PA-CF (PETG ok)	ABENICS housing: forearm sockets, post clamp, shoulders
54	90_center_ball	x1	PA-CF	Fixed centre ball (sphere bearing)
55	91_holder_retainer	x2	PETG	6808 outer-race retainer ring (3x M3 countersunk)
56	92_cradle_strap	x2	PETG	Theta-servo strap (closes the cradle, 2x M3)
57	93_mp_spacer	x2	PETG	Monopole pin spacer 10.5/8.2 x 6.5 (or brass tube)

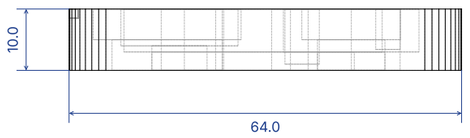
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x1    M3 insert pilot (depth 6.6)
  5.60  x1    M4 insert pilot (depth 9.0)
  5.95  x1    press fit 6 mm pin
  6.40  x1    M5 insert pilot (depth 10.5)
  7.95  x1    press fit 8 mm pin
  9.00  x1
  9.95  x1    press fit 10 mm pin
 11.00  x2
 12.20  x1    12 mm tube socket
 16.10  x1    bearing seat 688 / bearing seat 625
 18.00  x1
 19.10  x1    bearing seat 626 / bearing seat 6800
 22.10  x1    bearing seat 608
 24.10  x1    bearing seat 6802
 33.00  x1
 42.10  x1    bearing seat 6806
```

WRA-1 00_fit_test_coupon

PRINT FIRST: fit-test coupon (bearings, pins, inserts)

Material: same as production Qty: 1 Mass: 61 g est.

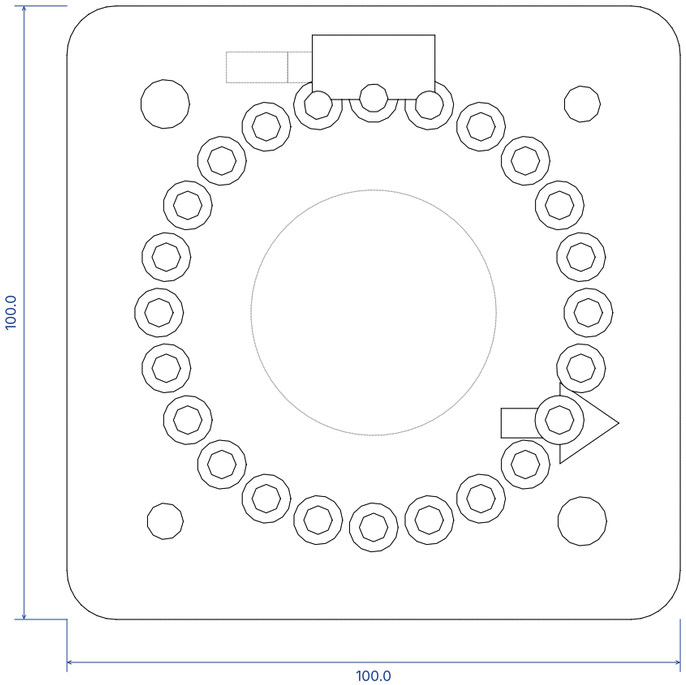
Print: 4 walls, 40% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

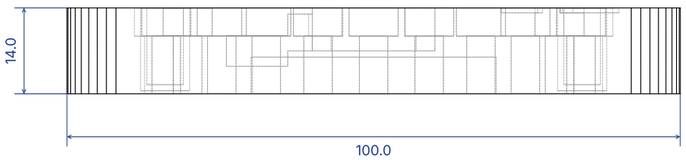
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 1/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

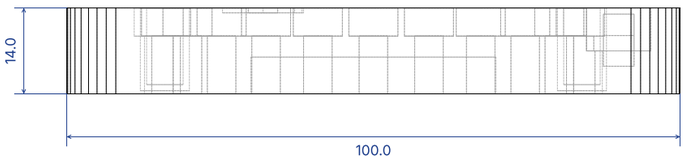
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.65  x24   M4 clearance
  5.95  x2    press fit 6 mm pin
  8.00  x26   M6 insert pilot (depth 13.5) / 626 inner-ring clearance
 40.00  x1
```

WRA-1 01_receiver

Dock receiver (chair side, same part L/R)

Material: PETG (PA-CF optional) Qty: 1 Mass: 104 g est.

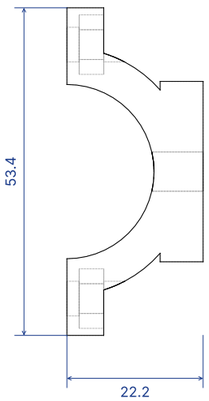
Print: 6 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

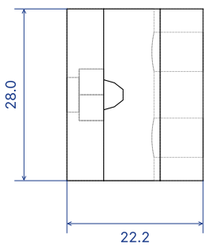
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 2/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

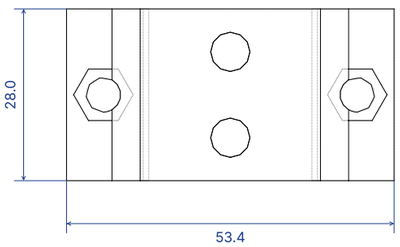
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  5.65  x2    M5 clearance
  6.40  x2    M5 insert pilot (depth 10.5)
 28.40  x1
```

WRA-1 02_tube_clamp_body

Tube clamp body (parametric tube OD)

Material: PA-CF recommended (PETG ok) Qty: 5 Mass: 15 g est.

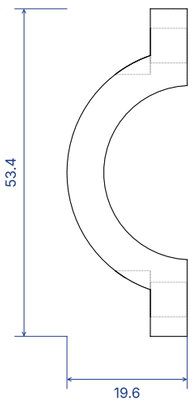
Print: 6 walls, 80% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

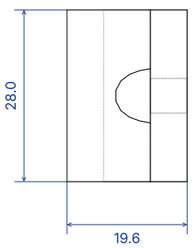
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 3/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

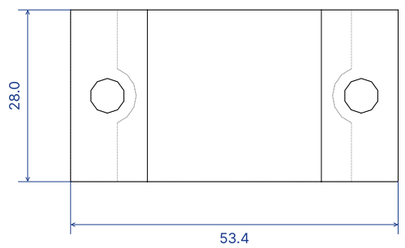
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  5.65  x2    M5 clearance
  9.40  x2
 28.40  x1
```

WRA-1 03_tube_clamp_cap

Tube clamp cap

Material: PA-CF recommended (PETG ok) Qty: 5 Mass: 12 g est.

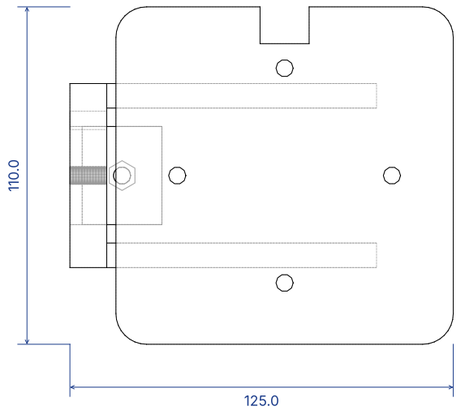
Print: 6 walls, 80% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

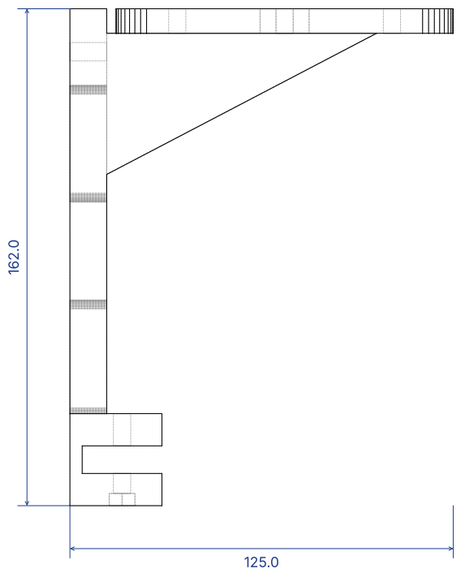
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 4/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

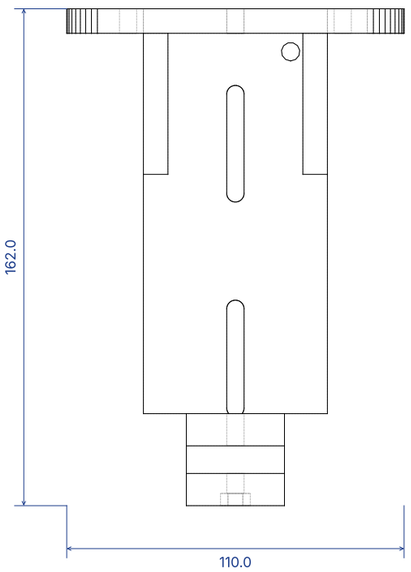
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  5.60  x4    M4 insert pilot (depth 9.0)
  5.65  x1    M5 clearance
  6.00  x1
```

WRA-1 04_post_bracket

Post bracket (spine + shelf), same part L/R

Material: PA-CF recommended Qty: 1 Mass: 211 g est.

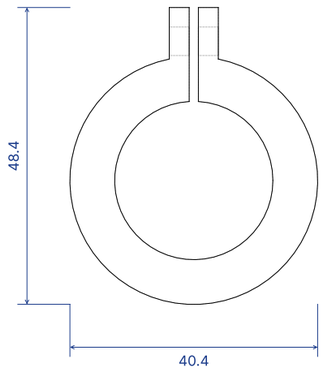
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:2 Units mm Third-angle projection Model (assembly) frame

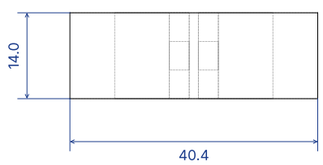
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 5/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

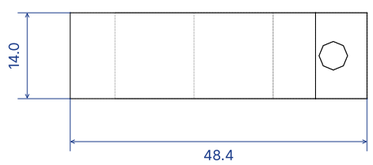
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.65  x1    M4 clearance
 25.80  x1
```

WRA-1 05_stop_collar

Post stop collar (secondary retention)

Material: PETG Qty: 1 Mass: 13 g est.

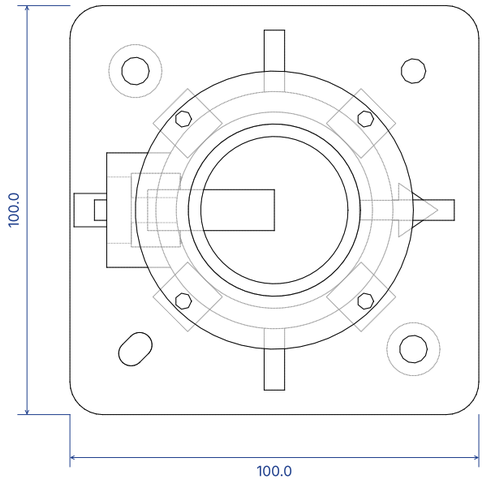
Print: 6 walls, 80% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

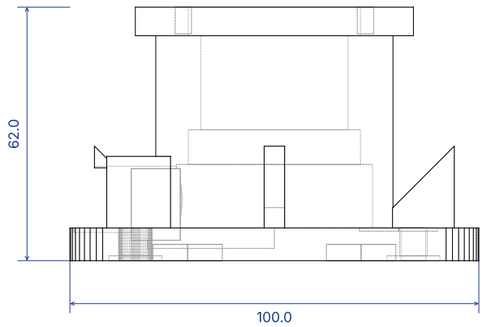
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 6/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

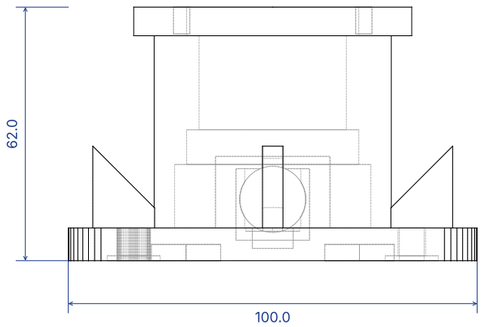
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
  6.05  x1
  6.75  x2    M6 clearance
 13.00  x2
 16.20  x1
 36.00  x1
 42.10  x1    bearing seat 6806
 48.00  x1
```

WRA-1 10_base_housing

Base housing + dock plate

Material: PA-CF (required: dock plate bending) Qty: 1 Mass: 122 g est.

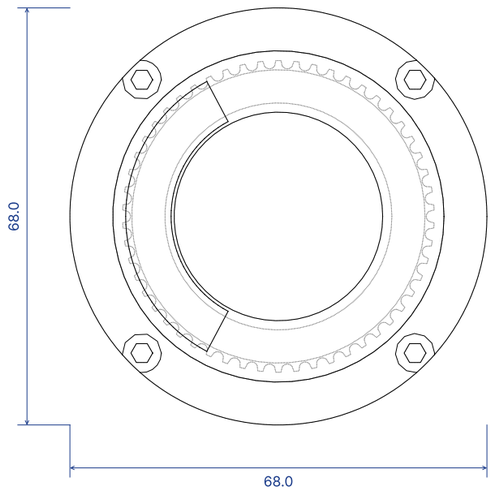
Print: 6 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

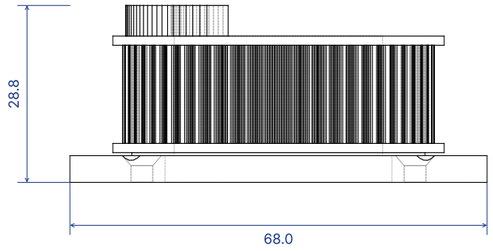
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 7/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

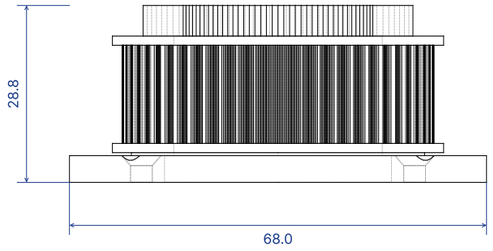
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
 34.00  x1
 37.00  x1
```

WRA-1 11_j1_fixed_pulley

J1 fixed pulley 54T (+ hard-stop arc)

Material: PETG Qty: 1 Mass: 42 g est.

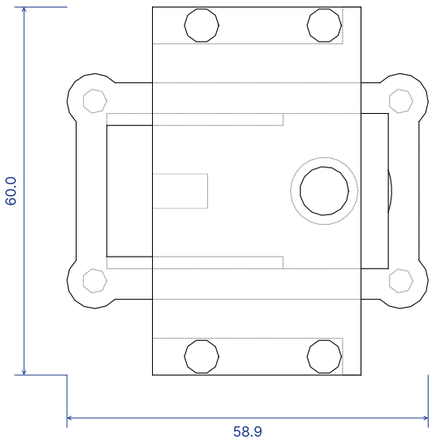
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

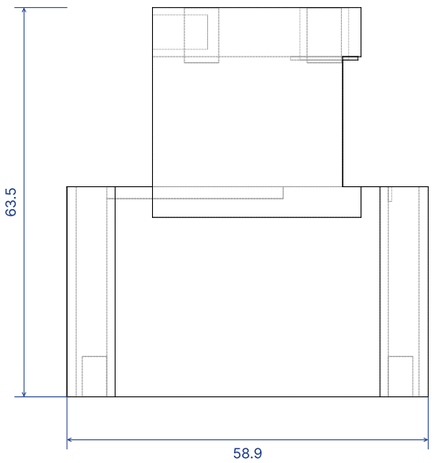
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 8/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

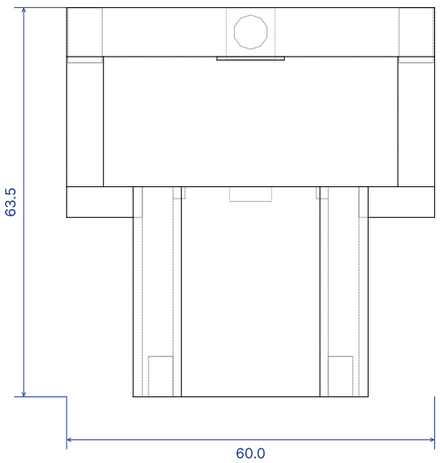
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
  5.60  x5    M4 insert pilot (depth 9.0)
  7.95  x1    press fit 8 mm pin
 22.00  x1
```

WRA-1 12_dm_bracket

Drive-module bracket

Material: PETG Qty: 1 Mass: 53 g est.

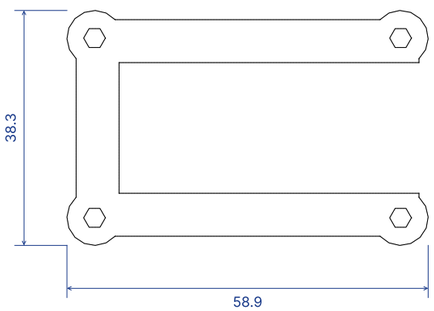
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

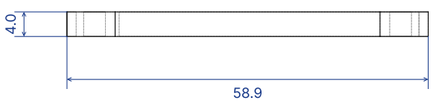
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 9/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

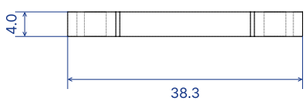
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
```

WRA-1 12_dm_cap

Drive-module servo cap

Material: PETG Qty: 3 Mass: 4 g est.

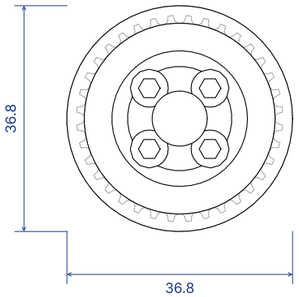
Print: 4 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

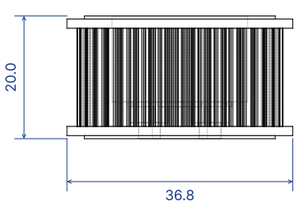
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 10/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

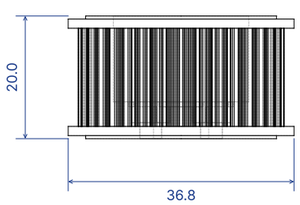
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  6.20  x4
  9.00  x1
 17.00  x1    6802 inner-ring clearance
 22.10  x1    bearing seat 608
```

WRA-1 12_dm_pulley36

36T HTD-3M drive pulley

Material: PETG Qty: 3 Mass: 14 g est.

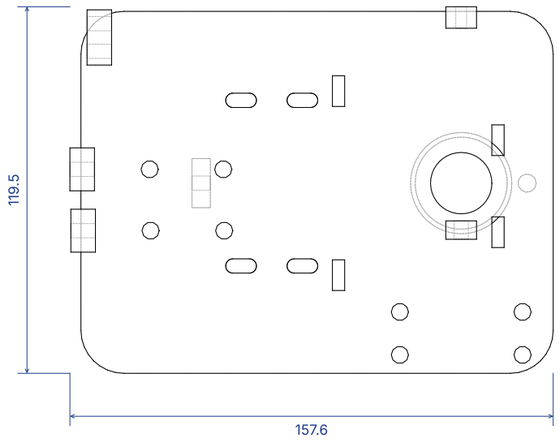
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

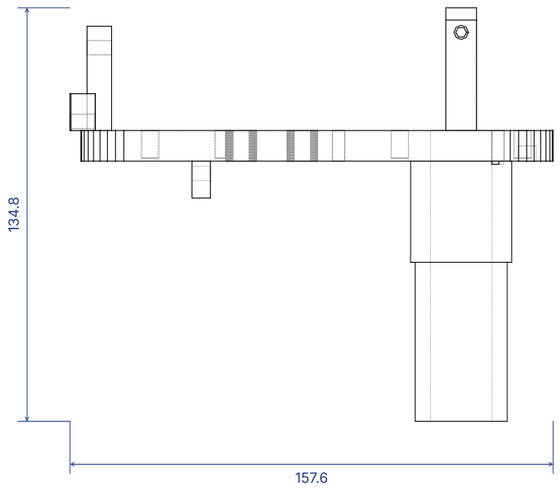
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 11/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

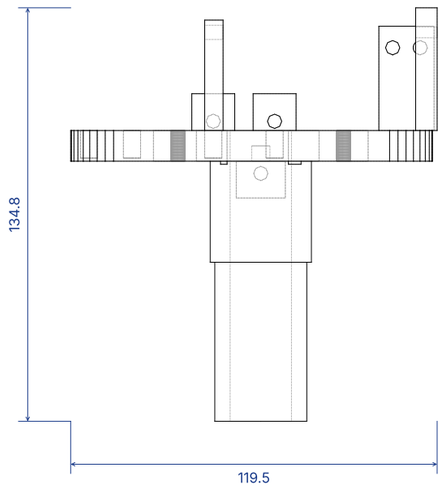
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x1    M3 clearance
  4.65  x6    M4 clearance
  5.60  x8    M4 insert pilot (depth 9.0)
  5.95  x1    press fit 6 mm pin
 20.00  x1
```

WRA-1 20_turntable

Turntable plate + J1 spindle

Material: PETG (PA-CF optional) Qty: 1 Mass: 179 g est.

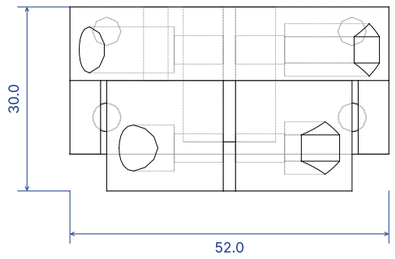
Print: 6 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:2 Units mm Third-angle projection Model (assembly) frame

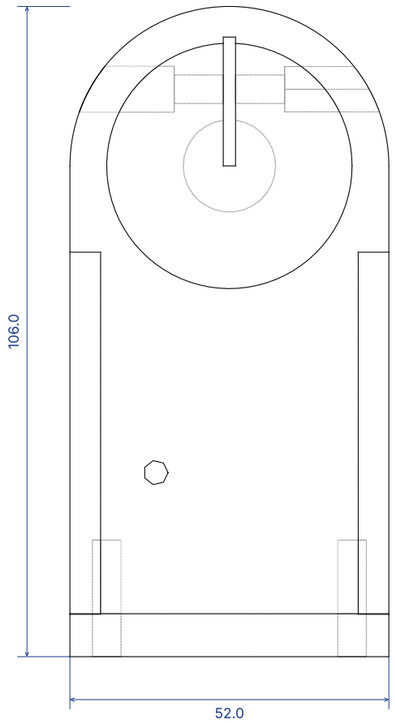
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 12/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

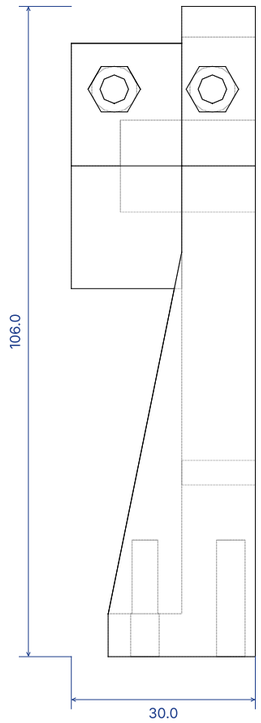
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x1    M3 insert pilot (depth 6.6)
  4.65  x6    M4 clearance
  7.50  x2
 15.05  x1
```

WRA-1 21_stub_cheek

Stub cheek (clamps J2 stub axle)

Material: PETG (PA-CF recommended) Qty: 1 Mass: 80 g est.

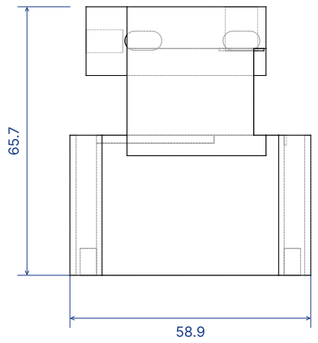
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

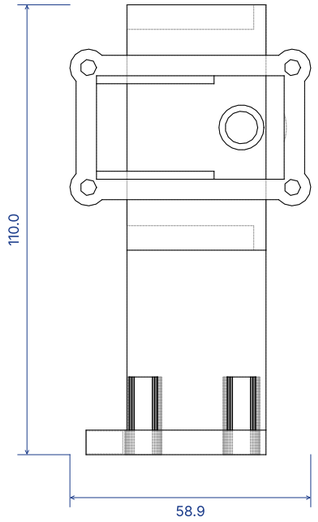
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 13/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

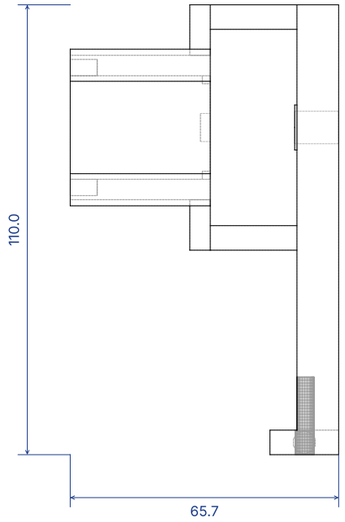
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
  5.60  x1    M4 insert pilot (depth 9.0)
  7.95  x1    press fit 8 mm pin
 22.00  x1
```

WRA-1 22_ddm_bracket_j2

Dual drive bracket (J2 half)

Material: PETG Qty: 1 Mass: 72 g est.

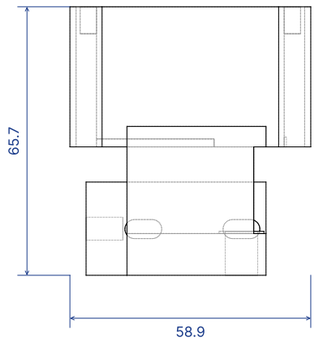
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

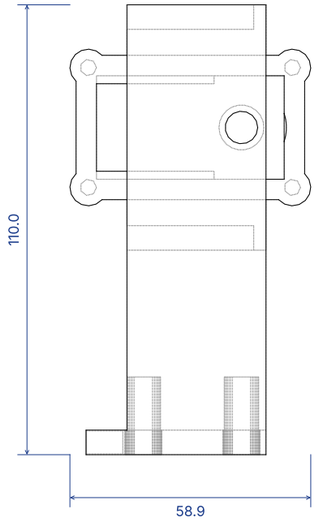
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 14/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

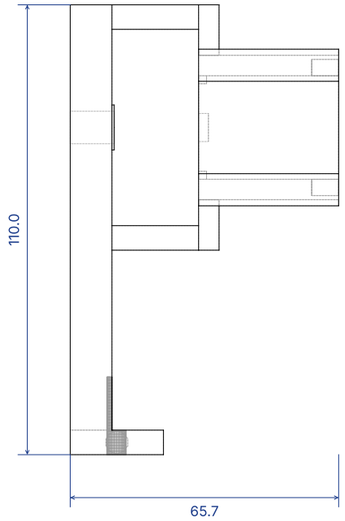
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
  5.60  x1    M4 insert pilot (depth 9.0)
  7.95  x1    press fit 8 mm pin
 22.00  x1
```

WRA-1 22_ddm_bracket_j3

Dual drive bracket (J3 half)

Material: PETG Qty: 1 Mass: 74 g est.

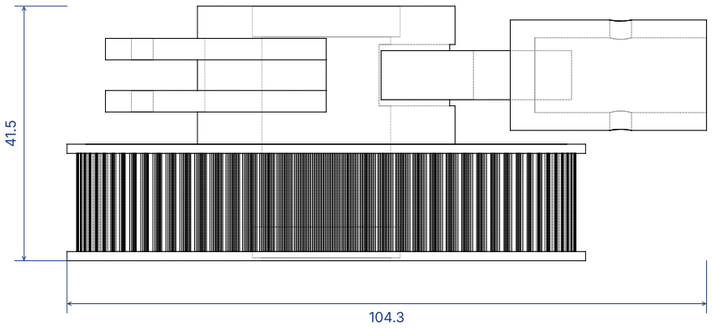
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

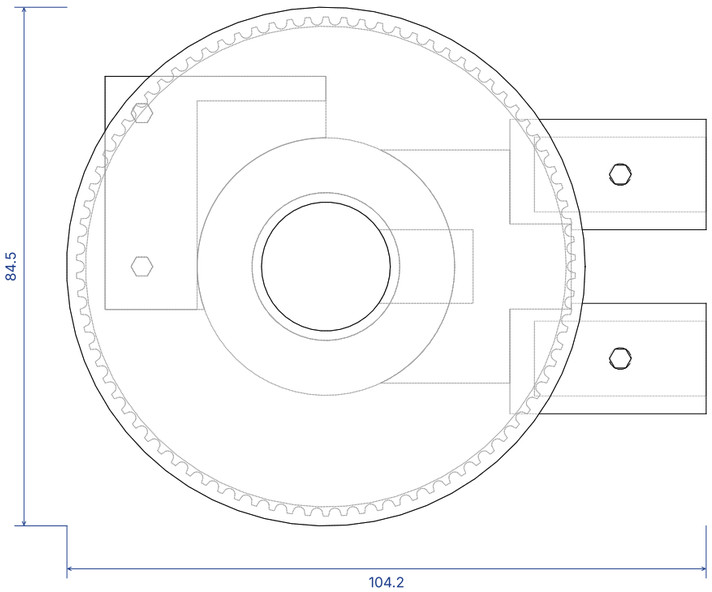
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 15/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

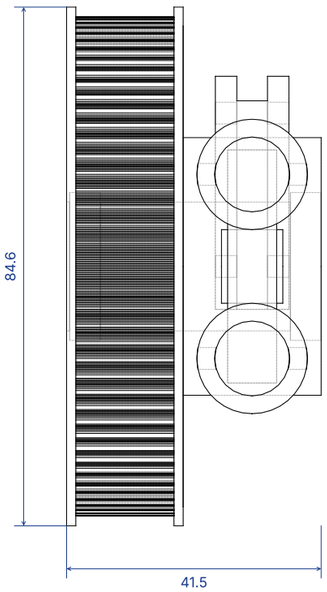
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
 12.20  x2    12 mm tube socket
 21.00  x1
 24.10  x1    bearing seat 6802
```

WRA-1 30_ua_hub

Upper-arm hub with integral 86T (J2) pulley

Material: PETG (PA-CF recommended) Qty: 1 Mass: 126 g est.

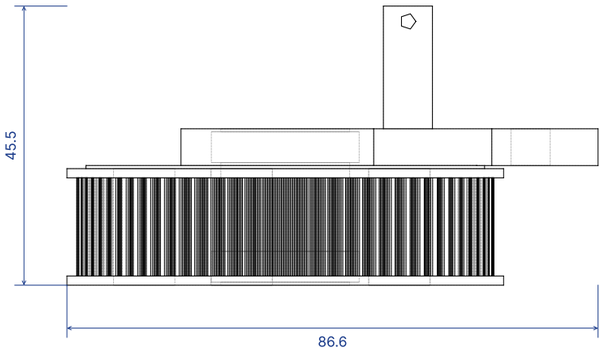
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

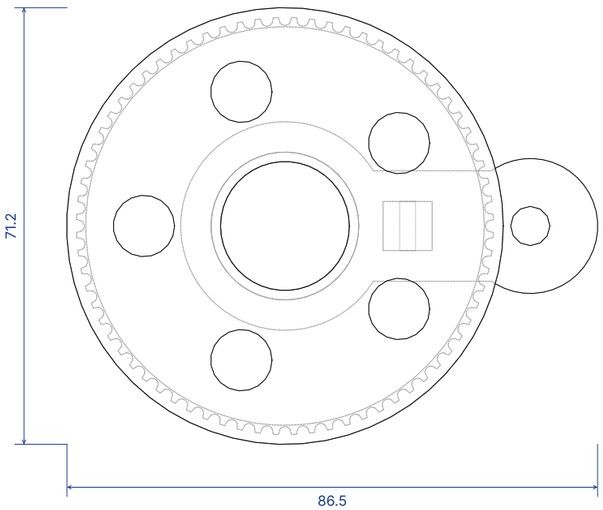
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 16/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

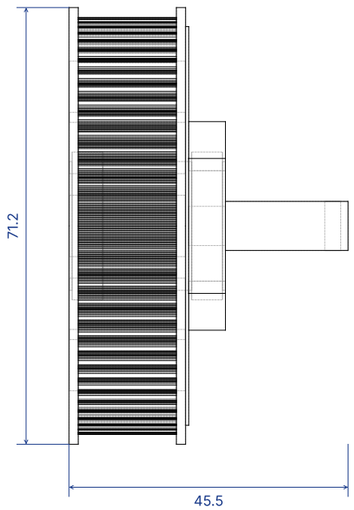
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.60  x1
  6.40  x1    M5 insert pilot (depth 10.5)
 10.00  x5    608 inner-ring clearance / 688 inner-ring clearance
 21.00  x1
 24.10  x1    bearing seat 6802
```

WRA-1 31_j3_hub

J3 hub: 72T pulley + parallelogram crank + spring L-lever

Material: PETG (PA-CF recommended) Qty: 1 Mass: 62 g est.

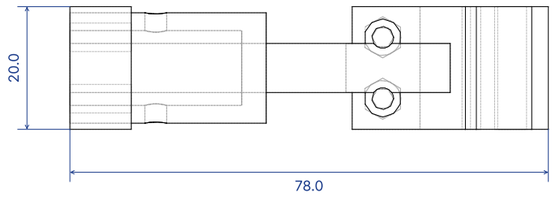
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

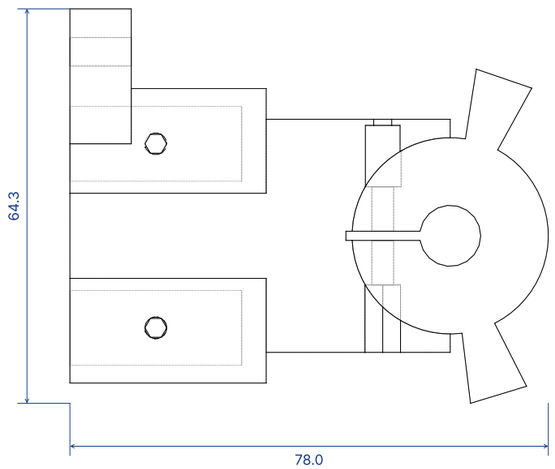
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 17/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

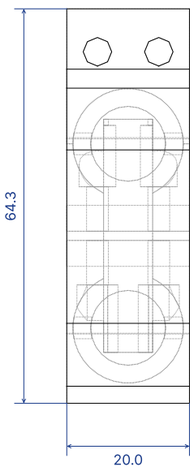
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  4.65  x2    M4 clearance
  6.00  x2
  9.95  x1    press fit 10 mm pin
 12.20  x2    12 mm tube socket
```

WRA-1 32_elbow_block

Elbow block (clamps elbow pin, J2 spring anchor, stop lugs)

Material: PETG (PA-CF recommended) Qty: 1 Mass: 40 g est.

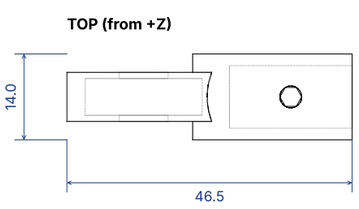
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

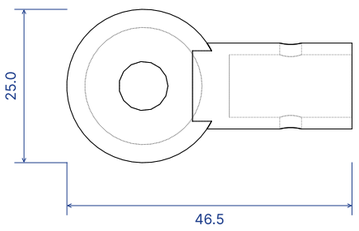
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 18/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

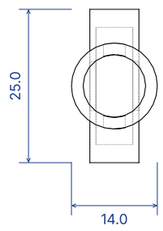
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x1    M3 clearance
  8.00  x1    M6 insert pilot (depth 13.5) / 626 inner-ring clearance
 10.20  x1    10 mm tube socket
 19.10  x1    bearing seat 626 / bearing seat 6800
```

WRA-1 33_rod_end

Parallelogram rod end (626 bearing)

Material: PETG Qty: 2 Mass: 5 g est.

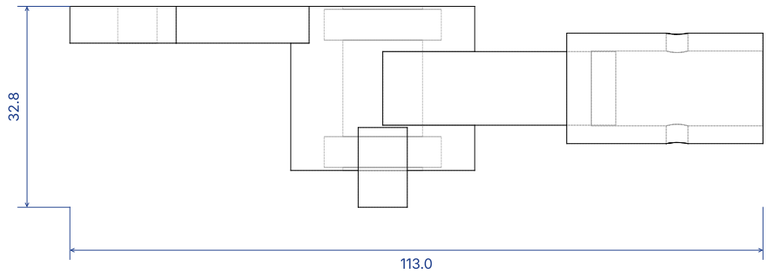
Print: 5 walls, 80% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

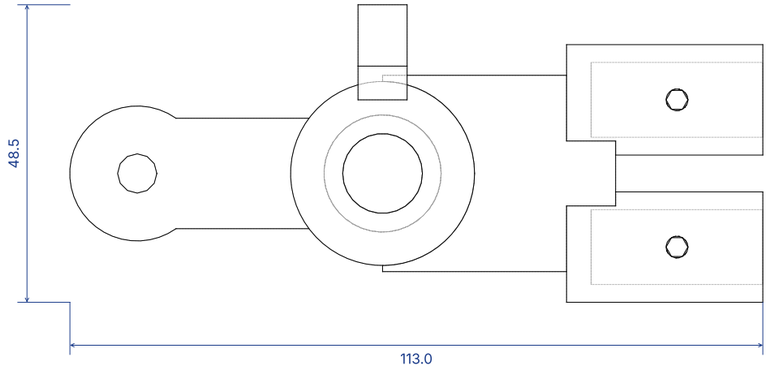
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 19/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

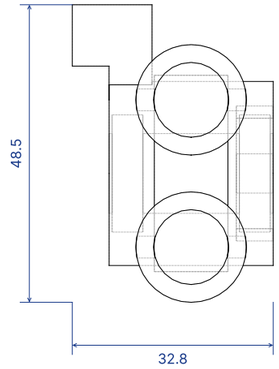
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x2    M3 clearance
  6.40  x1    M5 insert pilot (depth 10.5)
 12.20  x2    12 mm tube socket
 13.00  x1
 19.10  x1    bearing seat 626 / bearing seat 6800
```

WRA-1 34_forearm_root

Forearm root: elbow bearings + crank + stop tab

Material: PETG (PA-CF recommended) Qty: 1 Mass: 39 g est.

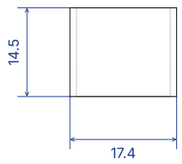
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

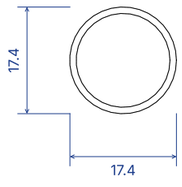
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 20/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

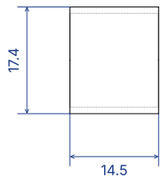
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
 15.30  x1
```

WRA-1 35_stub_spacer_14p46mm

Stub inner-ring spacer 14.46 mm

Material: PETG Qty: 1 Mass: 1 g est.

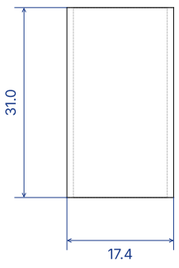
Print: 3 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

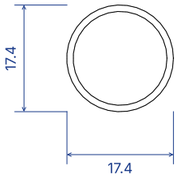
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 21/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

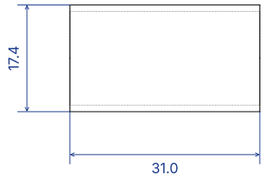
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
 15.30  x1
```

WRA-1 35_stub_spacer_30p96mm

Stub inner-ring spacer 30.96 mm

Material: PETG Qty: 1 Mass: 2 g est.

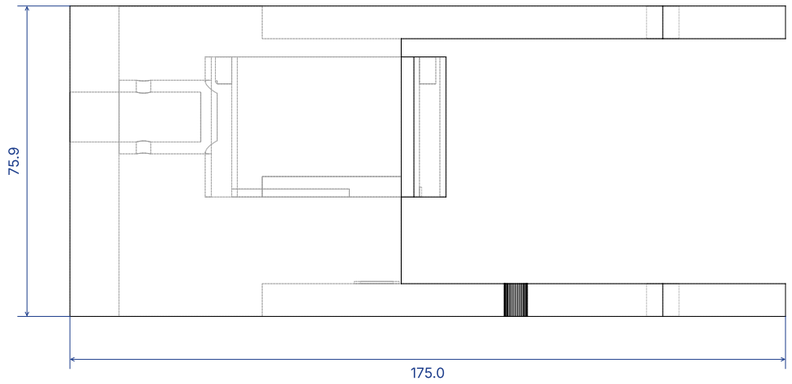
Print: 3 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

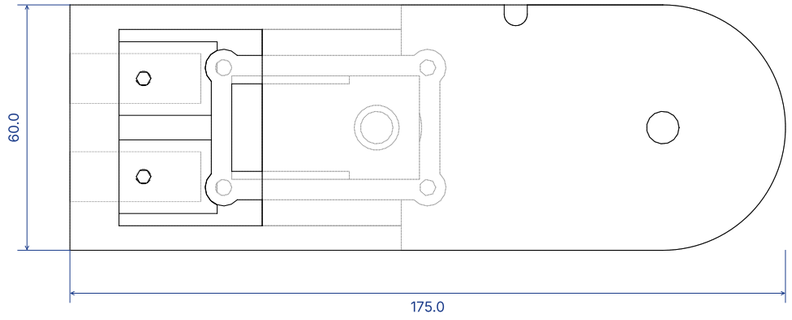
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 22/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

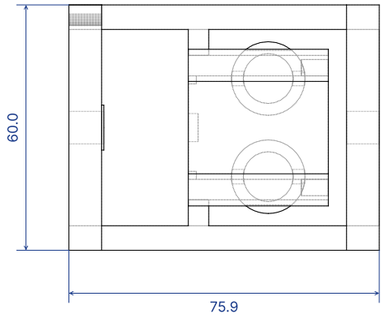
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x2    M3 clearance
  4.00  x4    M3 insert pilot (depth 6.6)
  7.95  x2    press fit 8 mm pin
 12.20  x2    12 mm tube socket
 22.00  x1
```

WRA-1 40_wrist_housing

Wrist housing: forearm sockets + J4 drive cradle + pitch fork

Material: PETG (PA-CF recommended) Qty: 1 Mass: 226 g est.

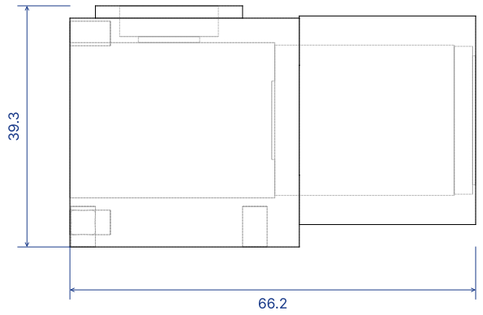
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

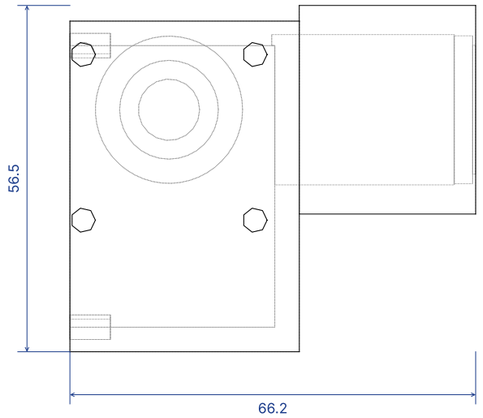
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 23/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

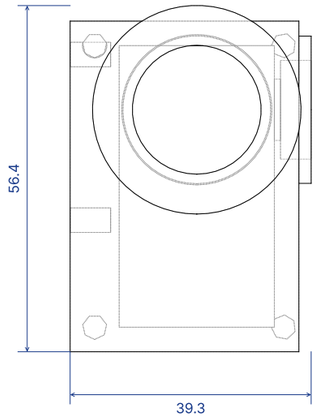
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x8    M3 insert pilot (depth 6.6)
 10.00  x1    608 inner-ring clearance / 688 inner-ring clearance
 16.10  x1    bearing seat 688 / bearing seat 625
 21.00  x1
 24.10  x1    bearing seat 6802
 24.50  x1
```

WRA-1 41_pitch_yoke

Pitch yoke: J5 servo pocket + roll bearing housing

Material: PETG (PA-CF recommended) Qty: 1 Mass: 45 g est.

Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

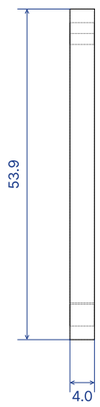
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 24/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

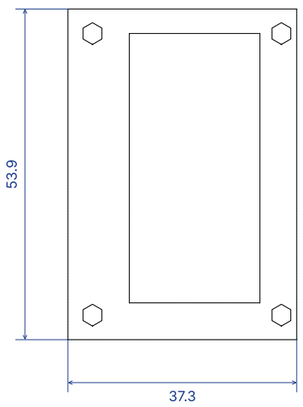
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
```

WRA-1 42_yoke_cap

Pitch-yoke rear cap

Material: PETG Qty: 1 Mass: 4 g est.

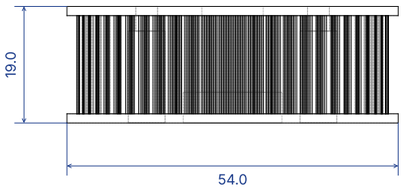
Print: 4 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

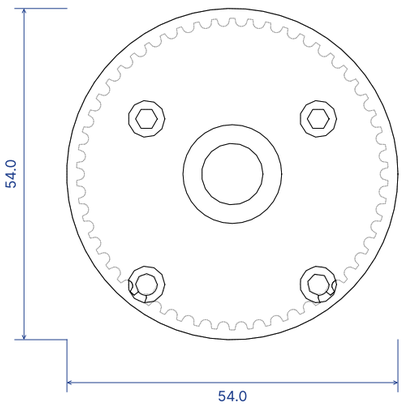
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 25/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

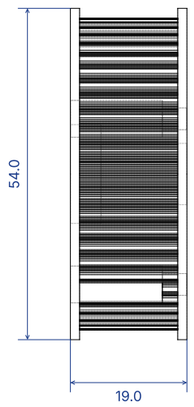
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  6.00  x4
 10.00  x1    608 inner-ring clearance / 688 inner-ring clearance
 16.10  x1    bearing seat 688 / bearing seat 625
```

WRA-1 43_j4_pulley54

J4 54T pulley (bolts to yoke)

Material: PETG Qty: 1 Mass: 43 g est.

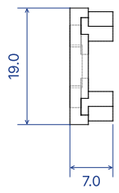
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

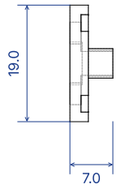
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 26/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

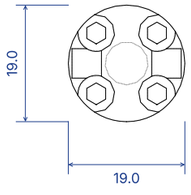
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  6.00  x4
  7.00  x1    625 inner-ring clearance
```

WRA-1 44_j5_horn_adapter

J5 horn adapter with drive dogs

Material: PETG (PA-CF) Qty: 1 Mass: 1 g est.

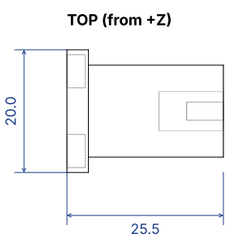
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

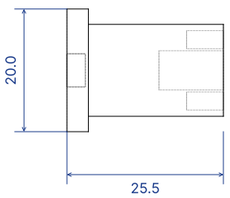
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 27/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

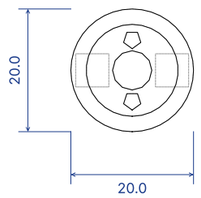
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.95  x2    press fit 3 mm pin
  6.40  x1    M5 insert pilot (depth 10.5)
```

WRA-1 45_roll_spindle

Roll spindle

Material: PA-CF or PETG Qty: 1 Mass: 5 g est.

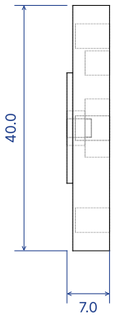
Print: 8 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

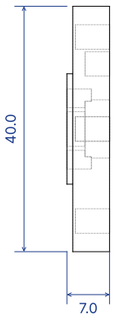
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 28/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

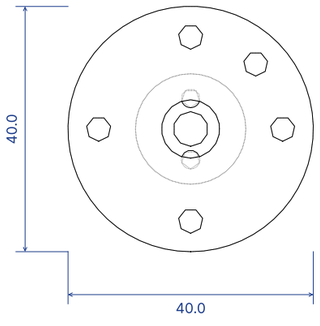
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.05  x2
  3.95  x1    press fit 4 mm pin
  4.00  x4    M3 insert pilot (depth 6.6)
  5.65  x1    M5 clearance
  9.50  x1
```

WRA-1 46_tool_flange

Tool flange (WRA-TF30 interface)

Material: PETG (PA-CF) Qty: 1 Mass: 9 g est.

Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

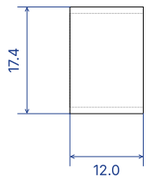
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 29/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

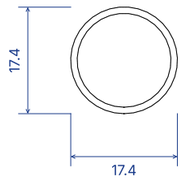
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
 15.30  x1
```

WRA-1 47_roll_spacer

Roll-spindle inner-ring spacer

Material: PETG Qty: 1 Mass: 1 g est.

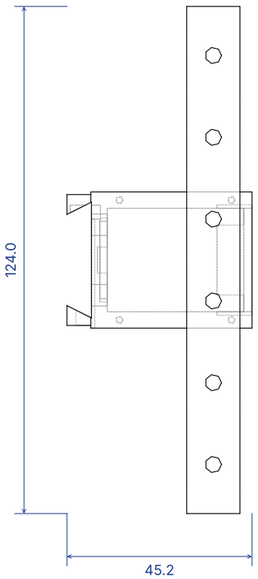
Print: 3 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

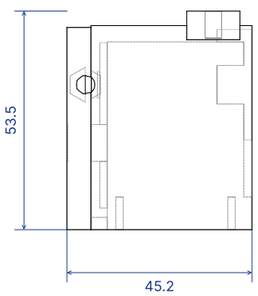
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 30/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

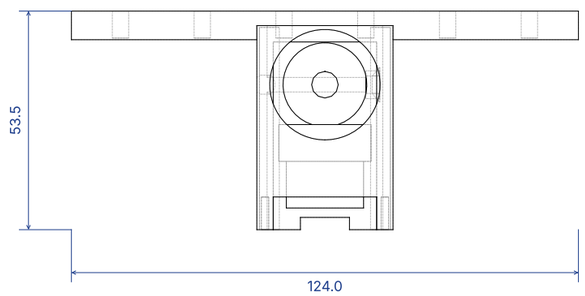
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  1.80  x4
  4.00  x6    M3 insert pilot (depth 6.6)
  4.65  x1    M4 clearance
  6.50  x1
 20.50  x1
 27.00  x1
```

WRA-1 50_claw_palm

Claw palm: dovetail, servo cradle, rail beam

Material: PA-CF (PETG ok) Qty: 1 Mass: 34 g est.

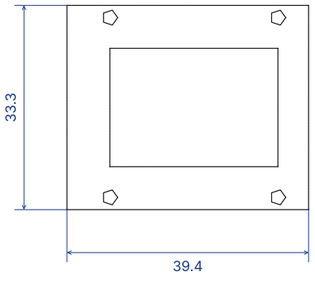
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

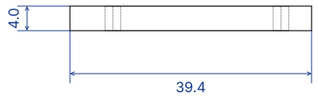
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 31/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

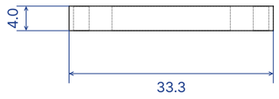
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.55  x4
```

WRA-1 51_claw_cradle_cap

Claw servo cradle cap

Material: PETG Qty: 1 Mass: 3 g est.

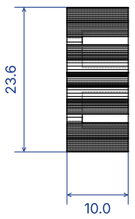
Print: 4 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

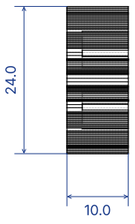
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 32/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

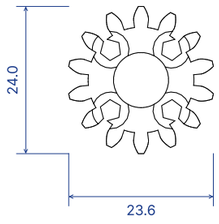
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  6.00  x4
  9.00  x1
```

WRA-1 52_claw_pinion

Claw pinion 14T m1.5 (on servo horn)

Material: PA-CF / PETG-CF Qty: 1 Mass: 2 g est.

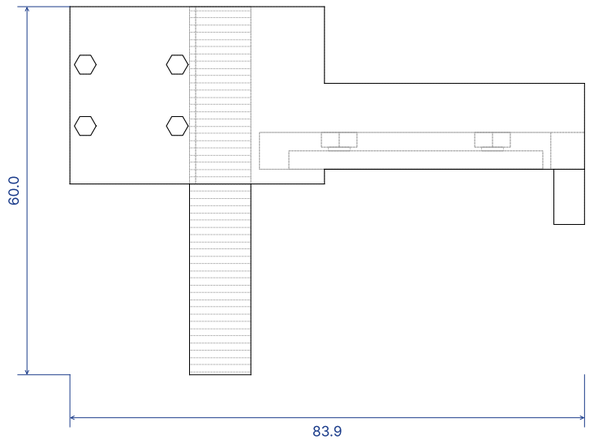
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

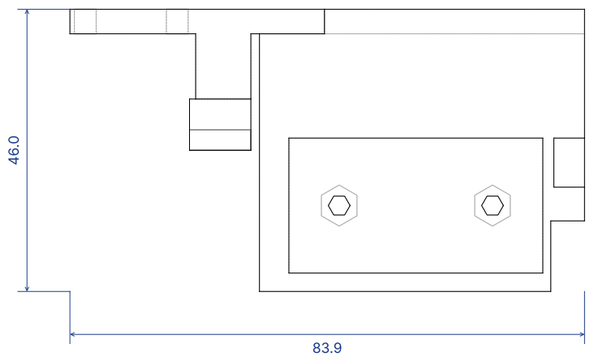
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 33/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

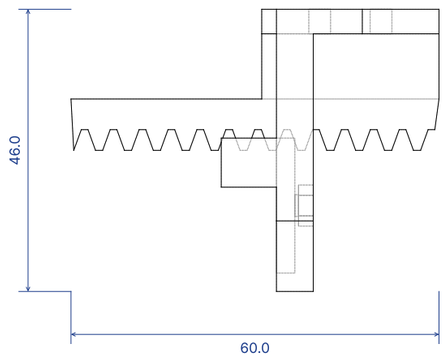
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x6    M3 clearance
```

WRA-1 53_claw_jaw_a

Claw jaw A (rack on top of pinion, upper knob lip)

Material: PA-CF / PETG-CF Qty: 1 Mass: 26 g est.

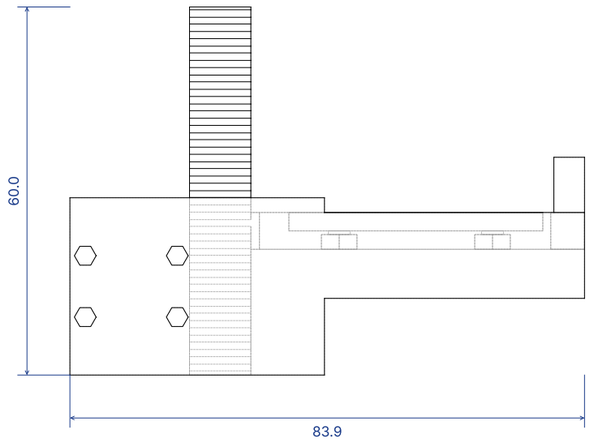
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

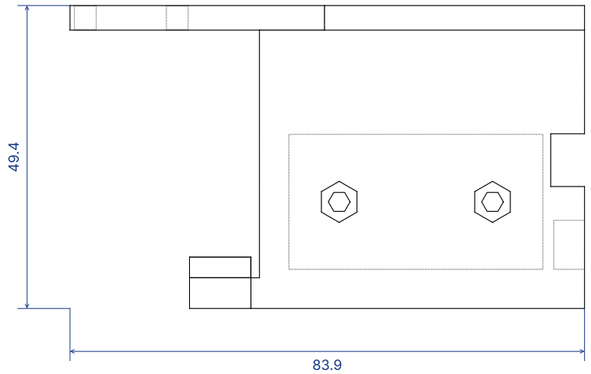
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 34/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

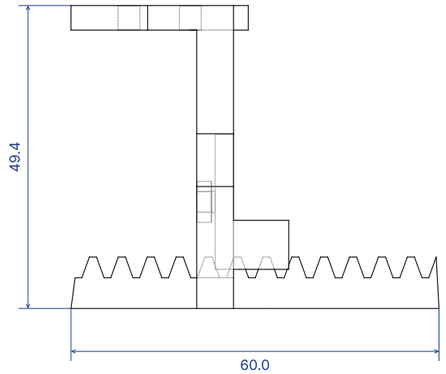
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x6    M3 clearance
```

WRA-1 53_claw_jaw_b

Claw jaw B (rack below pinion, lower knob lip)

Material: PA-CF / PETG-CF Qty: 1 Mass: 25 g est.

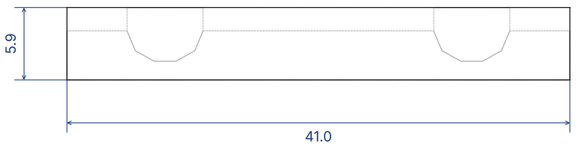
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

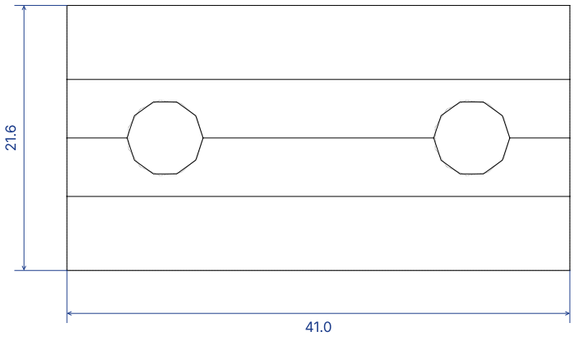
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 35/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

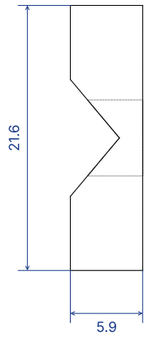
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  6.20  x2
```

WRA-1 54_claw_pad_a

Claw pad A, 100 deg V (glue 1.5 mm silicone facing)

Material: TPU 95A + silicone sheet Qty: 1 Mass: 3 g est.

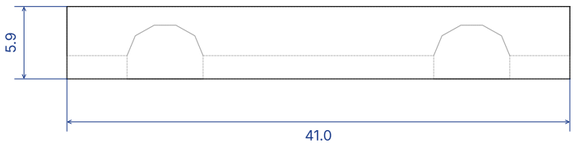
Print: 3 walls, 40% L/R-specific: no (same part both sides)

Scale 2:1 Units mm Third-angle projection Model (assembly) frame

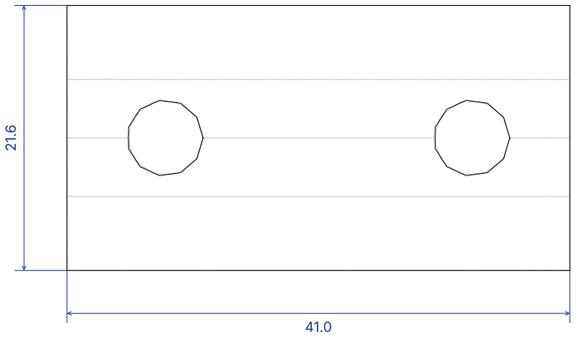
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 36/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

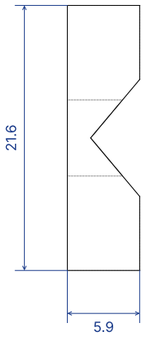
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  6.20  x2
```

WRA-1 54_claw_pad_b

Claw pad B (mirror of A)

Material: TPU 95A + silicone sheet Qty: 1 Mass: 3 g est.

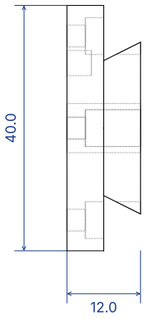
Print: 3 walls, 40% L/R-specific: no (same part both sides)

Scale 2:1 Units mm Third-angle projection Model (assembly) frame

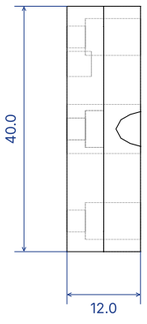
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 37/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

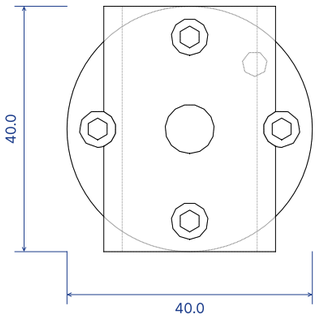
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  4.10  x1
  6.00  x4
  8.00  x1    M6 insert pilot (depth 13.5) / 626 inner-ring clearance
```

WRA-1 56_tool_adapter

Quick-change tool adapter (dovetail, both wrists)

Material: PA-CF (PETG ok) Qty: 1 Mass: 14 g est.

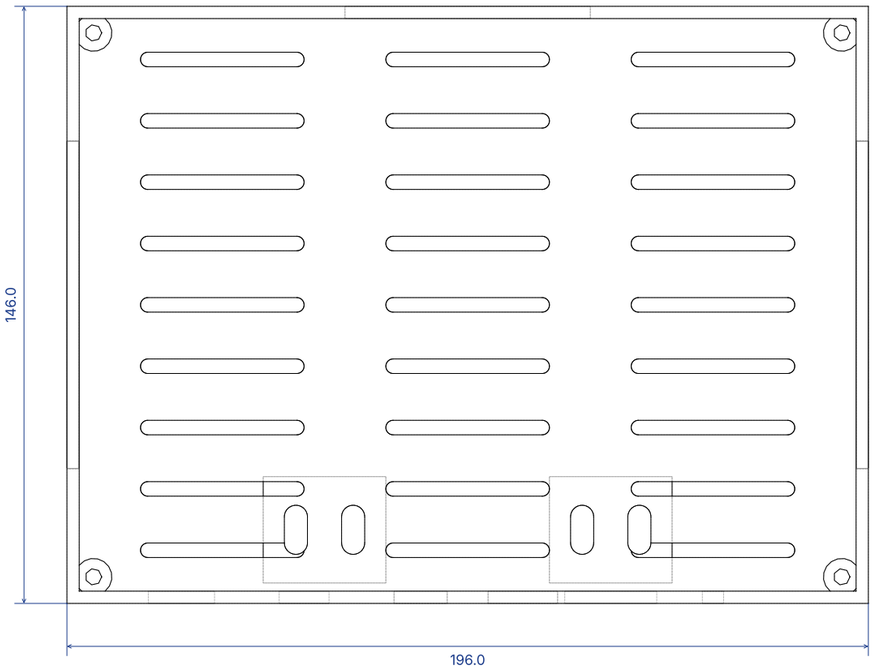
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

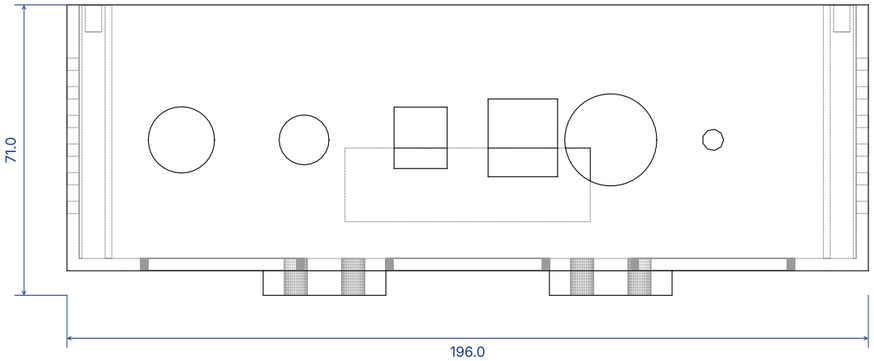
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 38/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

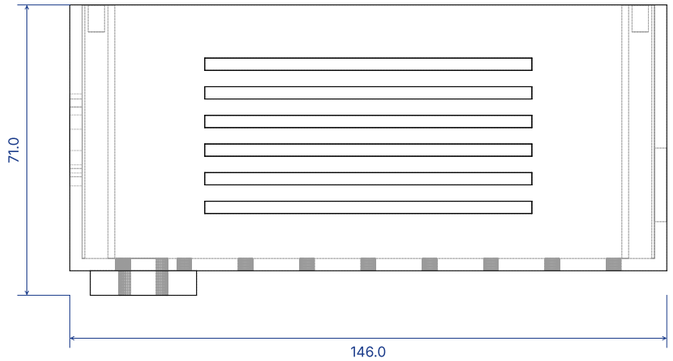
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
  5.20  x1
 12.20  x1    12 mm tube socket
 16.20  x1
 22.50  x1
```

WRA-1 60_electronics_box

Electronics enclosure (Jetson, ESP32, power, E-stop relays)

Material: PETG Qty: 1 Mass: 182 g est.

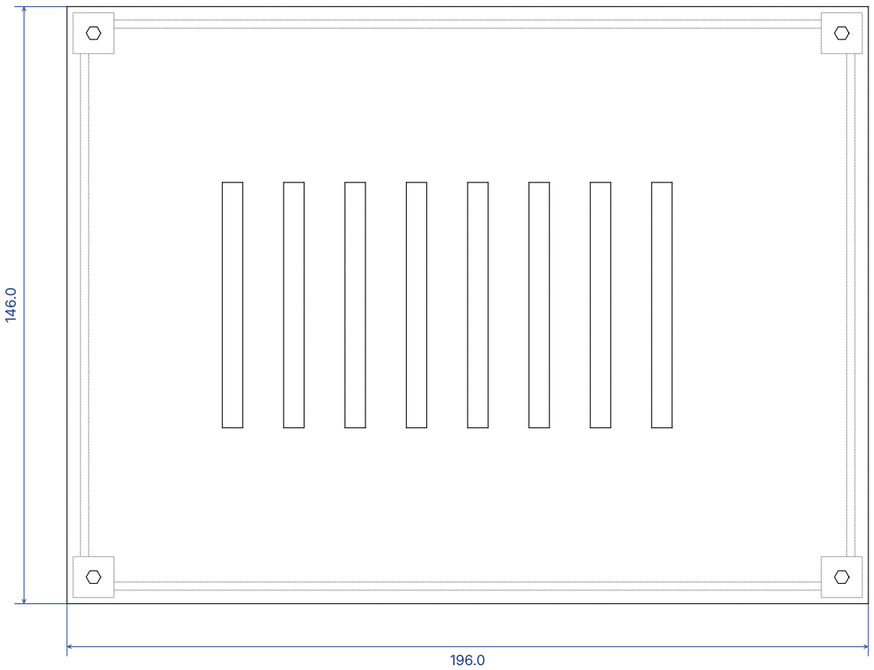
Print: 4 walls, 20% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

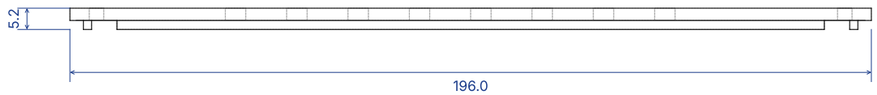
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 39/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

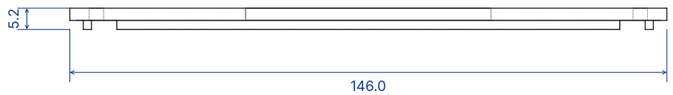
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
```

WRA-1 61_electronics_lid

Electronics enclosure lid

Material: PETG Qty: 1 Mass: 59 g est.

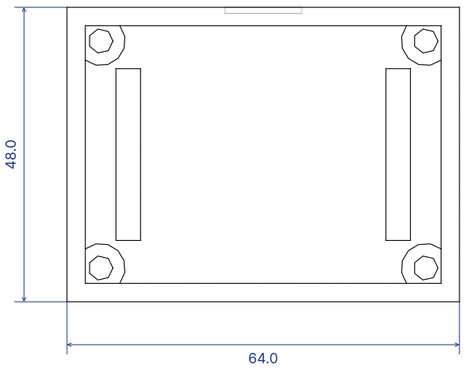
Print: 3 walls, 15% L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

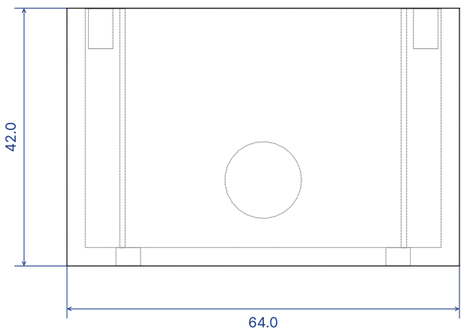
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 40/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

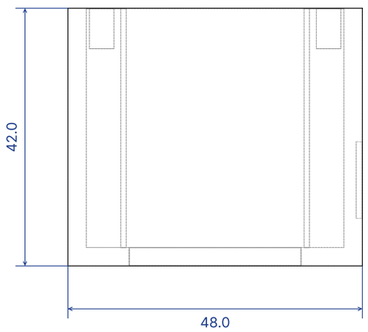
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  4.00  x4    M3 insert pilot (depth 6.6)
 12.50  x1
```

WRA-1 62_estop_pendant

E-stop pendant body

Material: PETG (yellow recommended) Qty: 1 Mass: 26 g est.

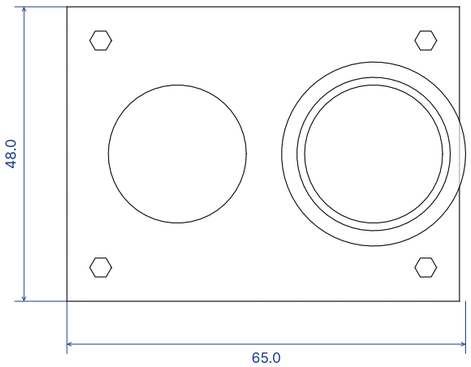
Print: 3 walls, 20% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

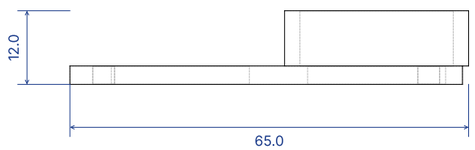
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 41/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

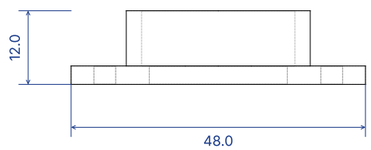
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
 22.50  x2
 25.00  x1
```

WRA-1 63_estop_pendant_lid

E-stop pendant lid (E-stop + guarded START)

Material: PETG (yellow) Qty: 1 Mass: 7 g est.

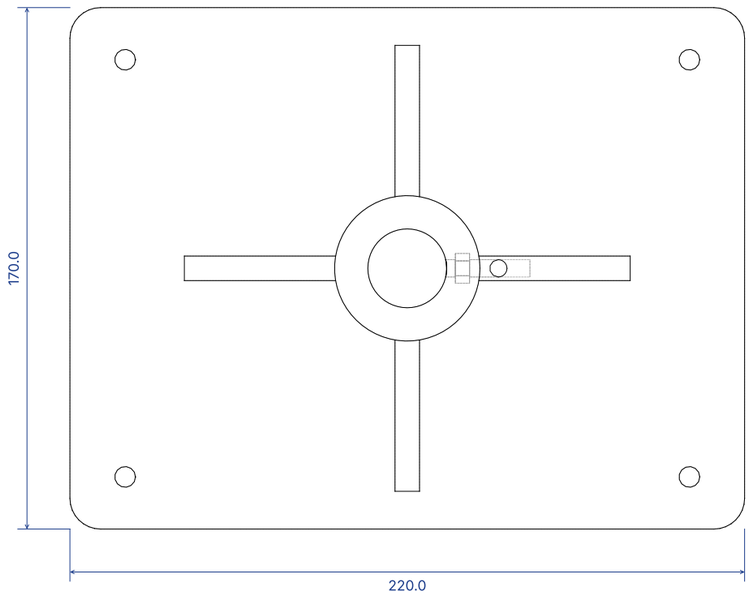
Print: 3 walls, 20% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

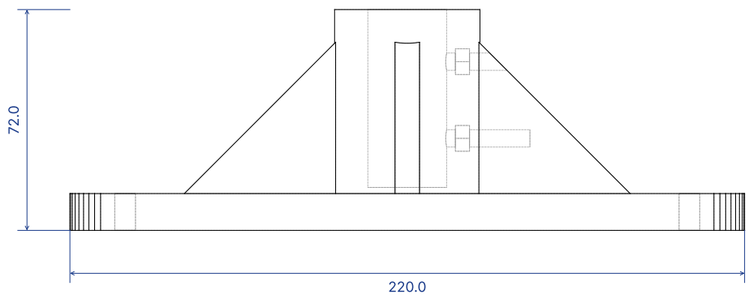
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 42/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

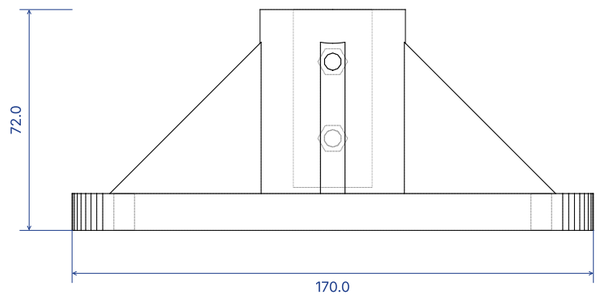
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  5.65  x2    M5 clearance
  6.75  x4    M6 clearance
 25.70  x1
```

WRA-1 70_bench_test_stand

Bench test stand (holds a post tube like the chair)

Material: PETG Qty: 1 Mass: 376 g est.

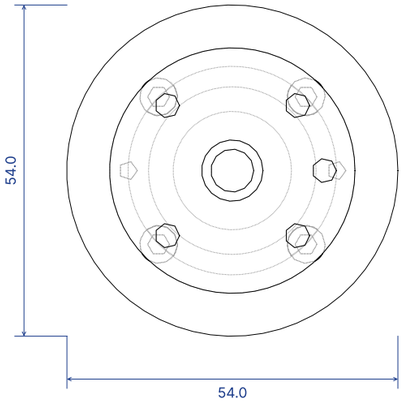
Print: 5 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:2 Units mm Third-angle projection Model (assembly) frame

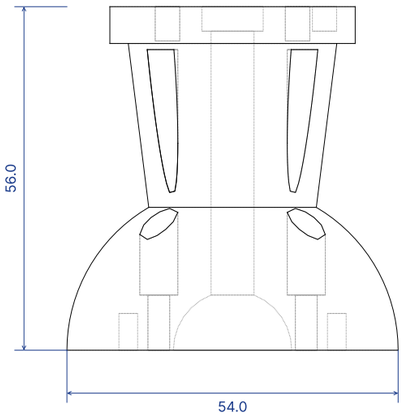
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 43/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

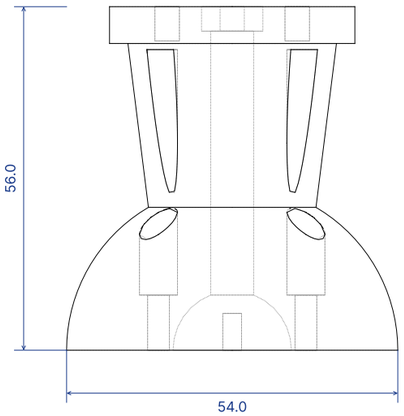
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.05  x2
  3.55  x4    M3 clearance
  3.95  x1    press fit 4 mm pin
  4.00  x4    M3 insert pilot (depth 6.6)
  6.20  x4
  7.00  x1    625 inner-ring clearance
 10.00  x1    608 inner-ring clearance / 688 inner-ring clearance
```

WRA-1 80_cs_gear_front

CS-gear front hemisphere + tool flange

Material: PA-CF (PETG-CF ok) Qty: 1 Mass: 64 g est.

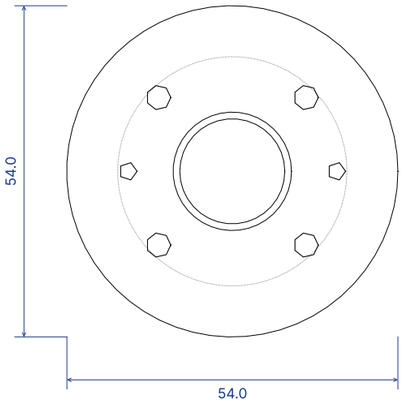
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

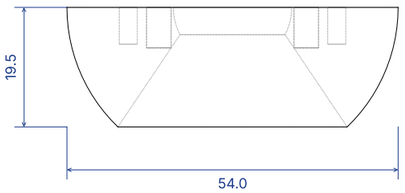
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 44/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

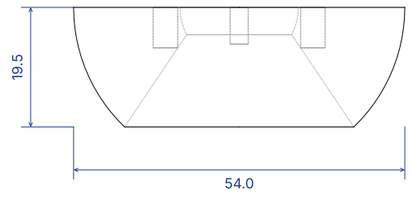
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.95  x2    press fit 3 mm pin
  4.00  x4    M3 insert pilot (depth 6.6)
```

WRA-1 81_cs_gear_back

CS-gear back hemisphere

Material: PA-CF (PETG-CF ok) Qty: 1 Mass: 22 g est.

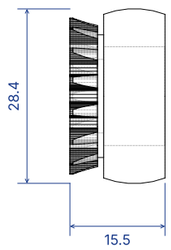
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

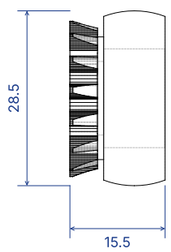
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 45/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

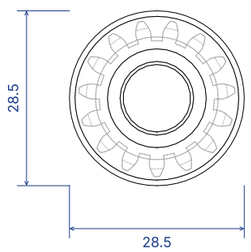
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
 11.00  x1
 12.00  x1    6800 inner-ring clearance
 16.10  x1    bearing seat 688 / bearing seat 625
```

WRA-1 82_monopole

Monopole S-gear 17T m1.5 + 15T m1.5 drive gear

Material: PA-CF Qty: 2 Mass: 5 g est.

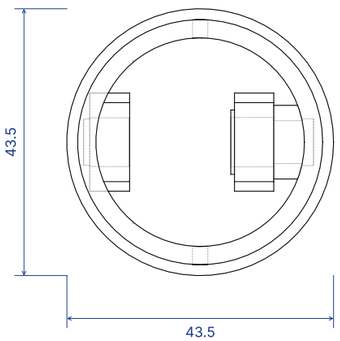
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

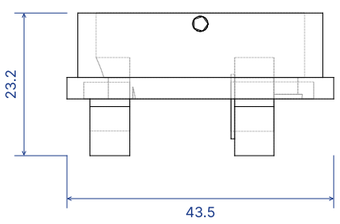
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 46/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

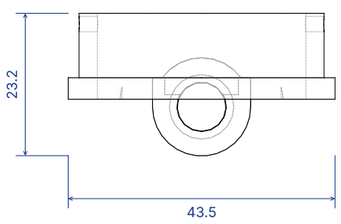
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.50  x1
  7.95  x1    press fit 8 mm pin
  8.05  x1
 26.70  x1
 34.00  x1
```

WRA-1 83_mp_fork

Monopole fork: pin arms + 6808 inner-race ring (carrier part 1)

Material: PA-CF (PETG-CF ok) Qty: 2 Mass: 9 g est.

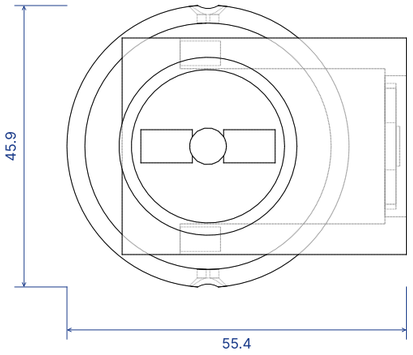
Print: 6 walls, 60% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

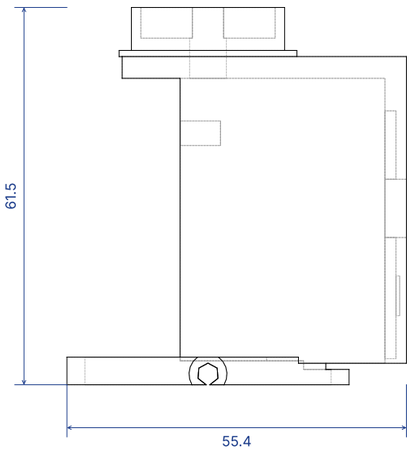
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 47/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

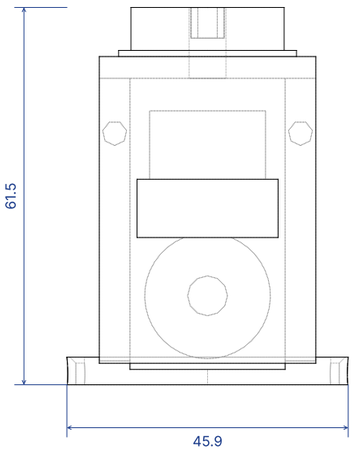
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x1    M3 clearance
  4.00  x2    M3 insert pilot (depth 6.6)
  6.00  x1
  6.50  x1
 20.50  x1
 40.15  x1
```

WRA-1 84_mp_cradle

Theta-servo cradle: sleeve, U-channel, 6805 hub, q dogs (carrier part

Material: PA-CF (PETG ok) Qty: 2 Mass: 29 g est.

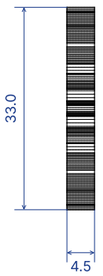
Print: 5 walls, 50% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

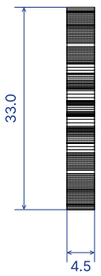
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 48/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

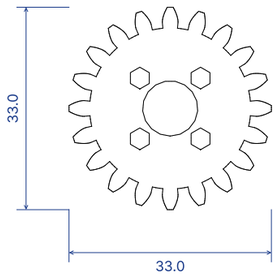
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  9.00  x1
```

WRA-1 85_theta_gear

Theta drive gear 20T m1.5 (on servo horn)

Material: PA-CF / PETG-CF Qty: 2 Mass: 3 g est.

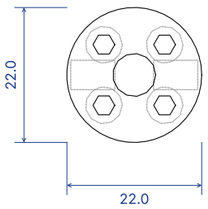
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

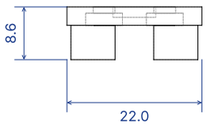
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 49/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

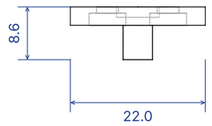
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
  6.00  x4
  7.00  x1    625 inner-ring clearance
```

WRA-1 86_q_coupling

q-servo horn coupling with drive dogs

Material: PA-CF / PETG Qty: 2 Mass: 1 g est.

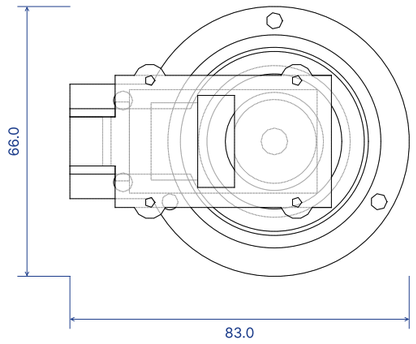
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

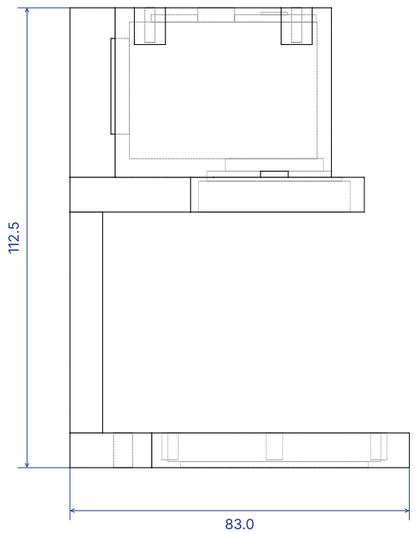
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 50/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

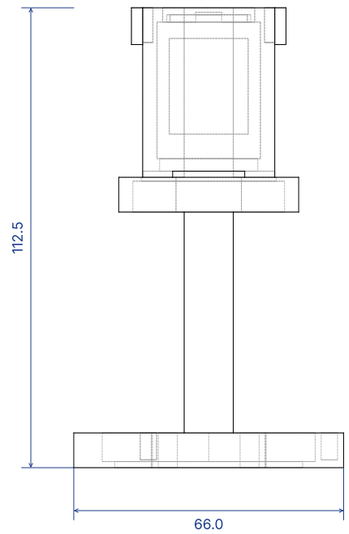
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.50  x4
  4.00  x3    M3 insert pilot (depth 6.6)
  4.65  x2    M4 clearance
  6.50  x1
 20.50  x1
 24.00  x1
 33.00  x1
 37.10  x1
 46.00  x1
 52.10  x1
```

WRA-1 87_module_frame

Module frame: 6808 holder, lug, strut, 6805 boss, q-servo cradle

Material: PA-CF (PETG ok) Qty: 2 Mass: 55 g est.

Print: 5 walls, 40% gyroid L/R-specific: no (same part both sides)

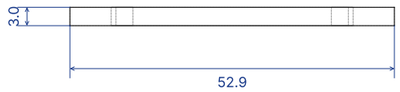
Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 51/57

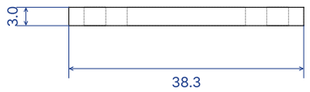
Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

TOP (from +Z)

FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x4    M3 clearance
```

WRA-1 88_q_servo_cap

q-servo cap

Material: PETG Qty: 2 Mass: 4 g est.

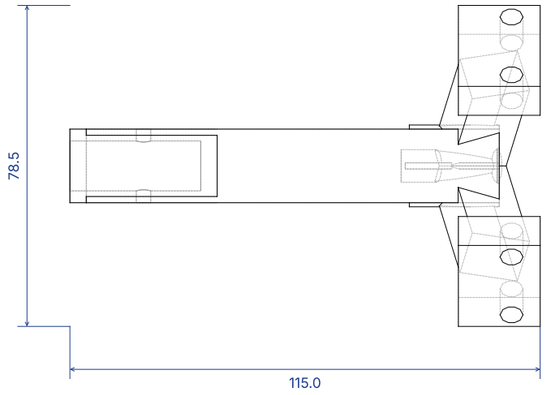
Print: 4 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

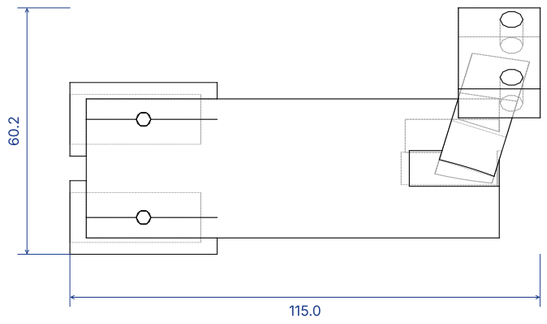
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 52/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

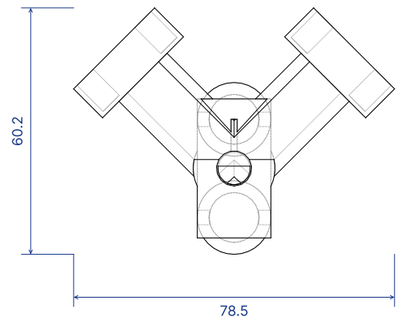
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x2    M3 clearance
  5.60  x4    M4 insert pilot (depth 9.0)
  7.95  x1    press fit 8 mm pin
  8.60  x1
 12.20  x2    12 mm tube socket
```

WRA-1 89_abenics_housing

ABENICS housing: forearm sockets, post clamp, shoulders

Material: PA-CF (PETG ok) Qty: 1 Mass: 55 g est.

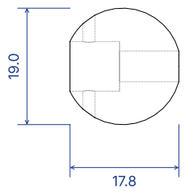
Print: 5 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1.5 Units mm Third-angle projection Model (assembly) frame

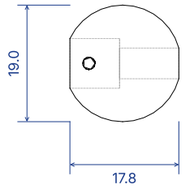
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 53/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

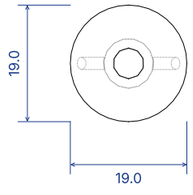
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  2.00  x1
  5.00  x1    623 inner-ring clearance
  8.10  x1
```

WRA-1 90_center_ball

Fixed centre ball (sphere bearing)

Material: PA-CF Qty: 1 Mass: 4 g est.

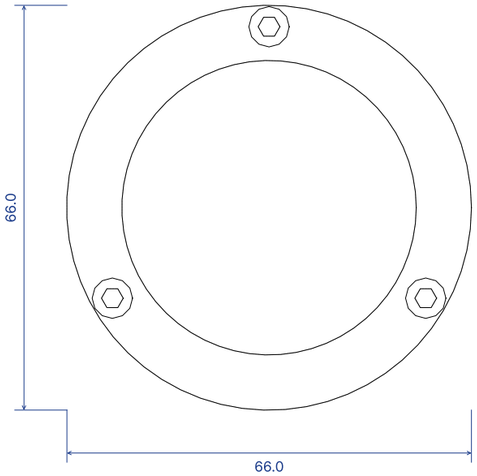
Print: 6 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

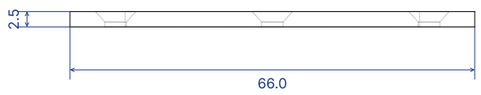
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 54/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

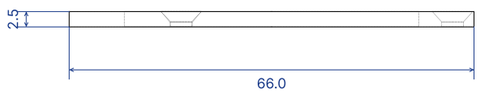
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x3    M3 clearance
 48.00  x1
```

WRA-1 91_holder_retainer

6808 outer-race retainer ring (3x M3 countersunk)

Material: PETG Qty: 2 Mass: 5 g est.

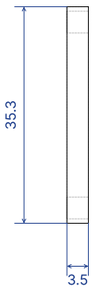
Print: 4 walls, 100% L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

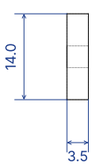
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 55/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

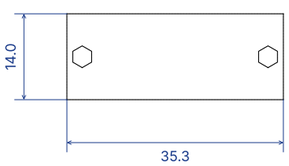
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  3.55  x2    M3 clearance
```

WRA-1 92_cradle_strap

Theta-servo strap (closes the cradle, 2x M3)

Material: PETG Qty: 2 Mass: 2 g est.

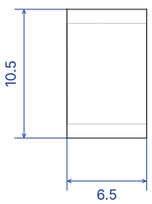
Print: 4 walls, 40% gyroid L/R-specific: no (same part both sides)

Scale 1:1 Units mm Third-angle projection Model (assembly) frame

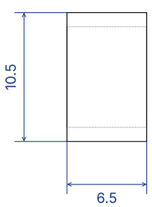
Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 56/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py

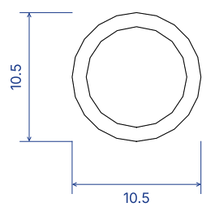
TOP (from +Z)



FRONT (from -Y)



RIGHT (from +X)




```
HOLE / BORE TABLE (auto-extracted, model nominal)
dia mm   qty   function
  8.20  x1
```

WRA-1 93_mp_spacer

Monopole pin spacer 10.5/8.2 × 6.5 (or brass tube)

Material: PETG Qty: 2 Mass: 0 g est.

Print: 3 walls, 100% L/R-specific: no (same part both sides)

Scale 2:1 Units mm Third-angle projection Model (assembly) frame

Fits: see print guide (bearing seats/pins/inserts from params.fit). Sheet 57/57

Generated 2026-10-12 from params.yaml sha256:b8b62fd5c287d452 - DO NOT EDIT, re-run build.py